Name: ____________________

Worksheet

Reading from graphs

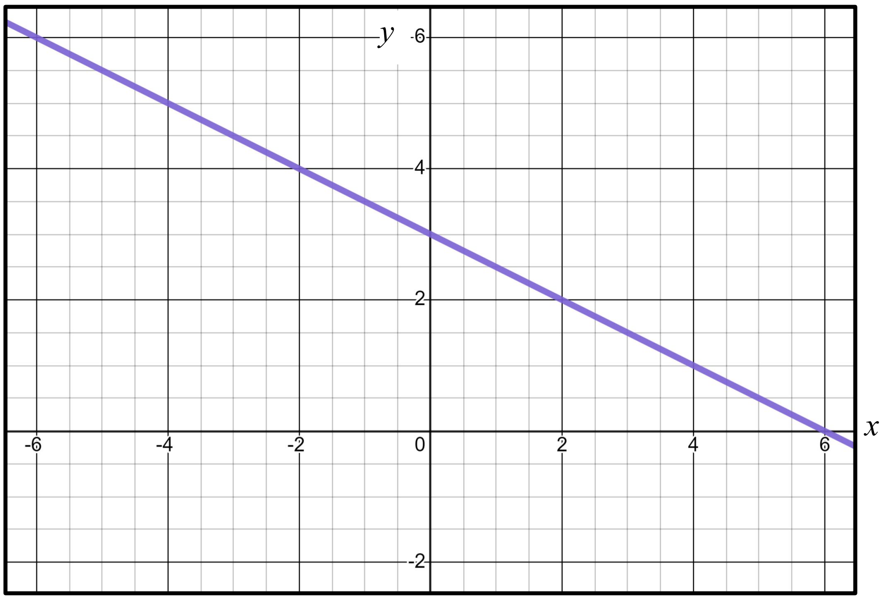
Task A: Reading from linear graphs

1) Use this graph of $y = 3 - \frac{1}{2}x$ to solve these equations.

a) $3 - \frac{1}{2}x = 1$

b) $3 - \frac{1}{2}x = 0$

c) $3 - \frac{1}{2}x = 5$

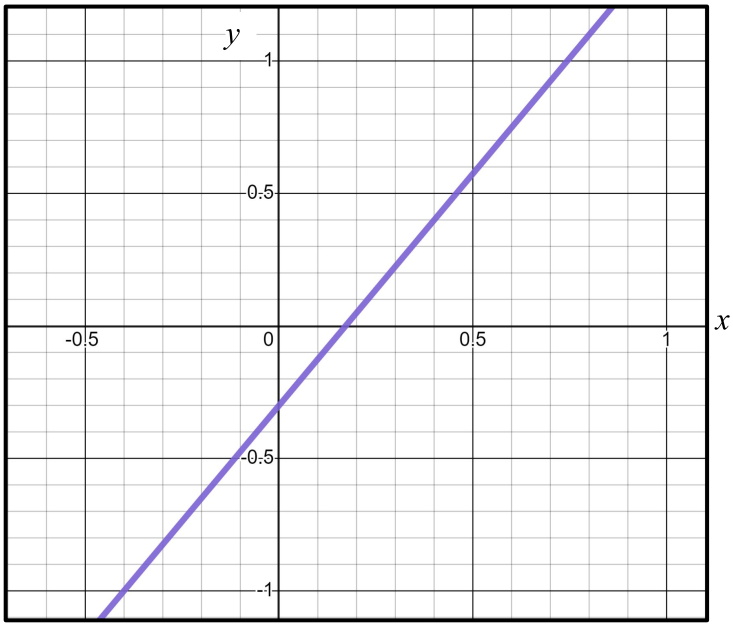


2) Use this graph of $y = 1.75x - 0.3$ to calculate:

a) $1.75 \times 0.8 - 0.3$

b) $1.75 \times 0.4 - 0.3$

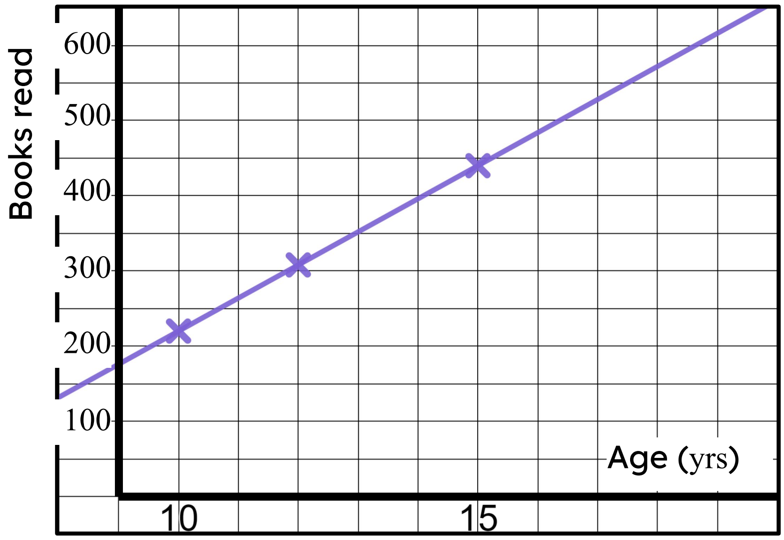
c) $1.75 \times -0.4 - 0.3$



3) The Oak pupils plot their age against how many books they have read and find a linear relationship.

a) Use the graph to estimate the age of a pupil who has read 400 books.

b) Use the graph to predict how many books a pupil will have read by the time they reach 18.



Task B: Reading from quadratic graphs

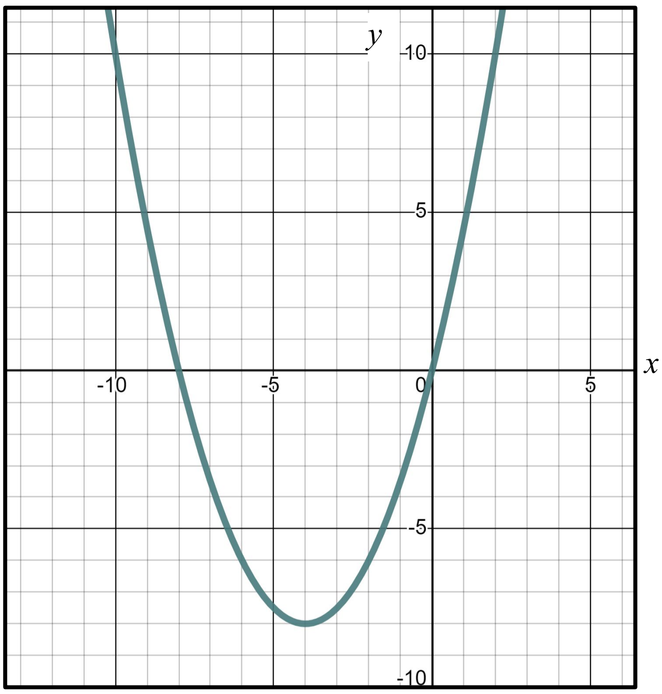
1) This is the graph of the quadratic equation $y = \frac{1}{2}x^2 + 4x$

Use the graph to solve:

a) $\frac{1}{2}x^2 + 4x = 10$

b) $\frac{1}{2}x^2 + 4x = 0$

c) $\frac{1}{2}x^2 + 4x = -8$



2) This parabola models the flight of a cricket ball being thrown.

Use the graph to read:

a) The height of the ball at a distance 18 metres.

b) The distance when the ball is at a height of 9 metres.

c) The distance when the ball is at a height of 12 metres.